MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 1102: INTRODUCTION TO PROGRAMMING

END OF SEMESTER EXAMINATION

DATE: MARCH, 2020 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

1. Answer **ALL FOUR** questions.
2. Marks for each question are indicated.
3. Write your answers clearly.
4. Indicate the question number for each answer.
5. This paper contains **TWO** pages excluding the cover page. Please check.

Question One

a. Define the following terms:

i. Local variable. **[2 Marks]**
ii. Global variable. **[2 Marks]**
iii. Pointer variable. **[2 Marks]**
iv. Parameter. **[2 Marks]**
v. Argument. **[2 Marks]**

b. Explain the following object-oriented programming concepts:

i. Inheritance. **[3 Marks]**
ii. Polymorphism. **[3 Marks]**
iii. Encapsulation. **[3 Marks]**
iv. Abstraction. **[3 Marks]**

Question Two

a. List **four** attributes of a good software tester. **[4 Marks]**

b. List **five** attributes of a good software. **[5 Marks]**

c. Explain **three** levels of software testing. **[6 Marks]**

d. Explain software testing life cycle. **[7 Marks]**

Question Three

a. Write pseudocode for the program that accepts odd integers in the range of 1 to 5 as input from the user, and outputs the equivalent of the number in words. If the user inputs a number that is not in the range, the output should be, "Invalid number".

i. Use IF…THEN…ELSE statement to solve this problem. **[10 Marks]**
ii. Use a CASE statement to solve this problem. **[10 Marks]**

b. Differentiate passing an argument by value from passing an argument by reference. **[6 Marks]**

c. Write a pseudocode for the program that passes an argument to a function by value. **[8 Marks]**

d. Write a pseudocode for the program that passes an argument to a function by reference. **[10 Marks]**

Question Four

An employee pays 15% of her salary as income tax to the government. Additionally, she pays 5% of her salary as medical scheme contribution. Write a pseudo code to calculate the amount paid to the government as her income tax and medical scheme contribution.

Note: Use a function called incomeTax to calculate the amount paid to the government and function called medicalScheme to calculate total medical scheme contribution.

[12 Marks]

END OF QUESTION PAPER

Program Start
USE VARIABLE